

Serial port control servo

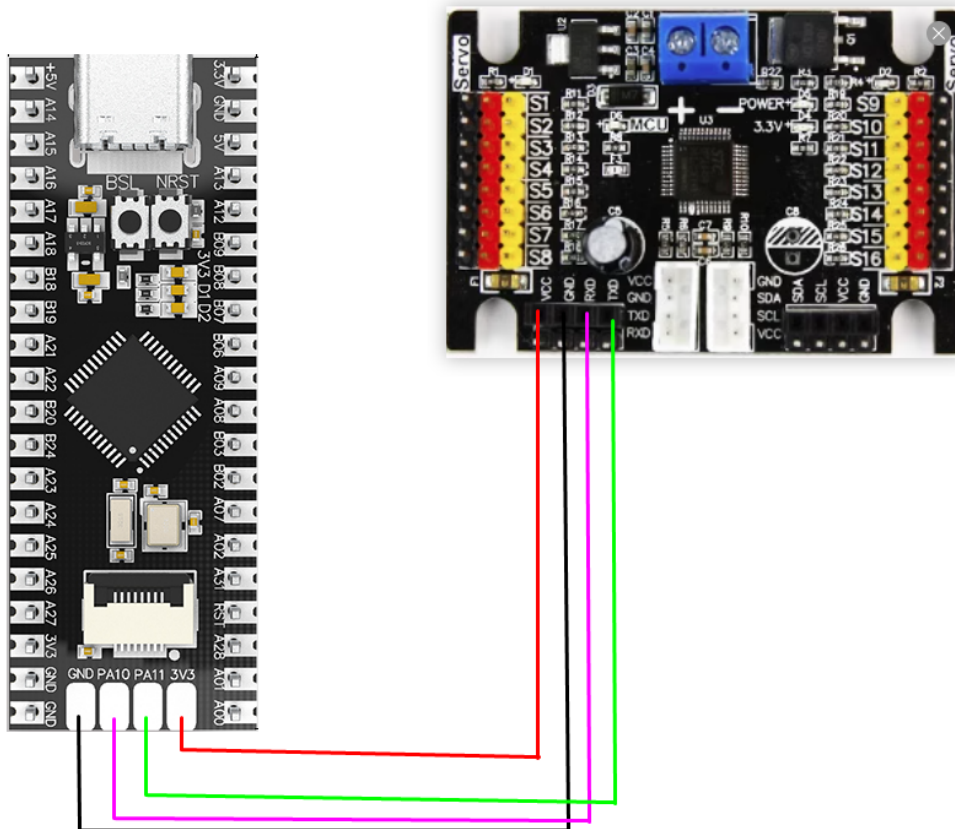
1. Learning objectives

Drive 180 degree servo through the serial port of 16-way servo driver board

2. Hardware connection

The servo used in this case is a 180 degree servo, which is connected to the S1 pin of the 16-way servo driver board

16-way servo driver board and MSPM0G3507 wiring



Note: The core board sold in this store has no solder pins at PA10 and PA11, which need to be soldered by yourself. Or change the serial port pin settings

3. Program description

- `usart.h`

```

#ifndef __USART_H__
#define __USART_H__

#include "ti_msp_dl_config.h"

void USART_Init(void);
static void USART_SendData(unsigned char data);
void UART_Servo(unsigned char servonum,unsigned char angle);

#endif

```

Header file that defines serial port transmission data and serial port control functions

- usart.c

```

void USART_Init(void)
{
    // SYSCFG初始化
    // SYSCFG initialization
    SYSCFG_DL_init();
    //清除串口中断标志
    //Clear the serial port interrupt flag
    NVIC_ClearPendingIRQ(UART_0_INST_INT_IRQN);
    //使能串口中断
    //Enable serial port interrupt
    NVIC_EnableIRQ(UART_0_INST_INT_IRQN);
}

//串口发送一个字节
//The serial port sends a byte
static void USART_SendData(unsigned char data)
{
    //当串口0忙的时候等待
    //wait when serial port 0 is busy
    while( DL_UART_isBusy(UART_0_INST) == true );
    //发送
    //send
    DL_UART_Main_transmitData(UART_0_INST, data);
}

void UART_Servo(unsigned char servonum,unsigned char angle)
{
    servonum = 64 + servonum;
    date1 = angle/100 + 48;
    date2 = (angle%100)/10 + 48;
    date3 = angle%10 + 48;
    USART_SendData(0x24); //发送包头 Sending packet header
    USART_SendData(servonum); //发送舵机编号 Send servo number
    USART_SendData(date1); //发送角度 Send Angle
    USART_SendData(date2); //发送角度 Send Angle
    USART_SendData(date3); //发送角度 Send Angle
}

```

```

    USART_SendData(0x23); //发送包尾 Sending packet tail
    delay_ms(100);
}

//串口的中断服务函数
//Serial port interrupt service function
void UART_0_INST_IRQHandler(void)
{
    uint8_t receivedData = 0;

    //如果产生了串口中断
    //If a serial port interrupt occurs
    switch( DL_UART_getPendingInterrupt(UART_0_INST) )
    {
        case DL_UART_IIDX_RX: //如果是接收中断 If it is a receive interrupt

            // 接收发送过来的数据保存 Receive and save the data sent
            receivedData = DL_UART_Main_receiveData(UART_0_INST);

            // 检查缓冲区是否已满 Check if the buffer is full
            if (recv0_length < RE_0_BUFF_LEN_MAX - 1)
            {
                recv0_buff[recv0_length++] = receivedData;
            }
            else
            {
                recv0_length = 0;
            }

            // 标记接收标志 Mark receiving flag
            recv0_flag = 1;

            break;

        default: //其他的串口中断 other serial port interrupts
            break;
    }
}

```

Define the serial port initialization function, the function of sending a byte of data, the serial port interrupt service function, and the serial port data sending function to communicate with the 16-channel servo driver board to control the servo.

The value range of servonum is (1-16), and the value range of angle is (0-180).

- empty.c

```

int main(void)
{
    int i=0;
    USART_Init();

    while (1)
    {
        for(i = 0;i<180;i+=5){
            UART_Servo(1,i);
        }
        UART_Servo(1,0);
    }
}

```

Assign values to the serial port control function. Use loops to control the continuous change of the servo from 0 to 180 degrees.

Note: The project source code must be placed in the SDK path for compilation.

For example, the path is: D:\TI\M0_SDK\mspm0_sdk_1_30_00_03\1.TB6612

新加卷 (D:) > TI > M0_SDK > mspm0_sdk_1_30_00_03				
名称	修改日期	类型	大小	
1.TB6612	2024/7/22 18:59	文件夹		
2.AT8236	2024/7/22 19:47	文件夹		
3.Encoder	2024/7/23 10:36	文件夹		
4.Servo	2024/7/23 11:13	文件夹		
docs	2024/7/23 10:33	文件夹		
examples	2024/7/23 10:34	文件夹		
kernel	2024/7/23 10:37	文件夹		
source	2024/7/23 10:33	文件夹		
tools	2024/7/23 10:33	文件夹		
imports.mak	2024/1/25 11:45	MAK 文件	2 KB	
known_issues_FAQ.html	2024/1/25 11:42	Microsoft Edge ...	67 KB	
license_mspm0_sdk_1_30_00_03.txt	2024/1/25 11:42	文本文档	33 KB	
manifest_mspm0_sdk_1_30_00_03.html	2024/1/25 11:42	Microsoft Edge ...	113 KB	
mspm0sdk_1_30_00_03.log	2024/7/23 10:42	文本文档	5,237 KB	
release_notes_mspm0_sdk_1_30_00_0...	2024/1/25 11:42	Microsoft Edge ...	108 KB	
uninstall.dat	2024/7/23 10:39	DAT 文件	344 KB	
uninstall.exe	2024/7/23 10:39	应用程序	6,048 KB	

4. Experimental phenomenon

Connect the servo to the S1 interface of the 16-channel servo driver board and burn the program to MSPM0G3507. After burning, connect the MSPM0G3507 to the 16-channel servo driver board according to the wiring diagram. After powering on, you will see the servo rotate from 0 to 180 degrees. Finally, return to 0 degrees, and repeat.